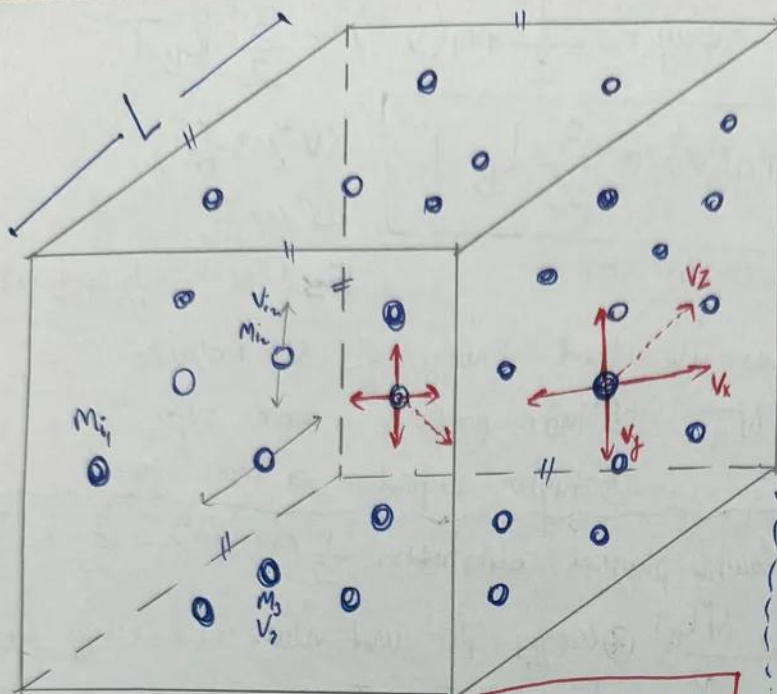


$$V = L^3$$

$$p = mv_x$$



$$\text{Boyle's Law} \Rightarrow P_1 V_1 = P_2 V_2$$

$$\text{Charles's Law} \Rightarrow V_1/T_1 = V_2/T_2$$

$$\text{Avogadro's Law} \Rightarrow V_1/n_1 = V_2/n_2$$

$$\text{Ideal Gas Law} \Rightarrow PV = nRT$$

$$\text{Momentum before hitting wall} = p = mv_x$$

$$\text{After bouncing back} = -mv_x$$

Change in momentum:

$$\Delta p = -mv_x - (mv_x) = -2mv_x$$

To travel across the box and return:

$$\Delta t = \frac{2L}{v_x}$$

Force = rate of momentum change:

$$F = \frac{\Delta p}{\Delta t} = \frac{-2mv_x}{\frac{2L}{v_x}} = -\frac{mv_x^2}{L}$$

this is force from one particle on one wall.

(1) Derivation of pressure from kinetic motion

$$P = \frac{1}{3} \frac{Nm \langle v^2 \rangle}{V}$$

Total force from all particles:

$$\text{isotropic motion: } \langle v^2 \rangle = \langle v_x^2 \rangle + \langle v_y^2 \rangle + \langle v_z^2 \rangle = 3 \langle v_x^2 \rangle \Rightarrow \langle v_x^2 \rangle = \frac{1}{3} \langle v^2 \rangle$$

Total force on one wall:

$$F_{\text{total}} = \sum_{i=1}^N \frac{mv_{x_i}^2}{L} = \frac{m}{L} \sum v_x^2 = \frac{Nm \langle v_x^2 \rangle}{L} = \frac{Nm}{L} \cdot \frac{1}{3} \langle v^2 \rangle$$

$$F = \frac{1}{3} \frac{Nm \langle v^2 \rangle}{L}$$

Pressure = Force / Area

$$P = \frac{F}{A} = \frac{1}{3} \cdot \frac{Nm \langle v^2 \rangle}{L} \cdot \frac{1}{L^2} = \frac{1}{3} \cdot \frac{Nm \langle v^2 \rangle}{L^3} = \frac{1}{3} \frac{Nm \langle v^2 \rangle}{V}$$

$$P = \frac{1}{3} \frac{Nm \langle v^2 \rangle}{V}$$

$$2) \text{Temperature from kinetic energy: } \frac{1}{2} m \langle v^2 \rangle = \frac{3}{2} k_B T$$

* Temperature is a measure of average kinetic energy per particle

* Energy comes from random motion, not from interaction or vibration (in ideal gas)

* If $T=0$, then $\langle v^2 \rangle = 0$: no motion

(1)

(2) Temperature from kinetic energy: $\frac{1}{2} m \langle v^2 \rangle = \frac{3}{2} k_B T$

$$P = \frac{1}{3} \frac{Nm \langle v^2 \rangle}{V}$$

$$\boxed{\frac{1}{2} m \langle v^2 \rangle = \frac{3}{2} k_B T}$$

$$\langle v^2 \rangle \propto \frac{T}{m}$$

$$\langle v^2 \rangle \sim \frac{T}{m}$$

physical calibration

$$E \sim \frac{1}{2} Nm \cdot \frac{T}{m} = \frac{1}{2} NT \Rightarrow \boxed{E \propto NT}$$

$$PV = N k_B T$$

Multiplying both sides by N gives the total Energy of All molecules

$$E = \frac{1}{2} Nm \langle v^2 \rangle$$

$$E \propto NT$$

⊛ More particles → more energy

⊛ higher temperature → more energy

Deriving (2) without using ideal gas law by physical calibration $\left(\frac{1}{2} m \langle v^2 \rangle = \frac{3}{2} k_B T \right)$ and deriving gas law

$$E \propto NT$$

$$E = N k_B T$$

$$\frac{E}{V} \sim \frac{N k_B T}{V}$$

⊛ energy per unit volume (i.e. energy density) is:

$$(E = \frac{3}{2} N k_B T)$$

$$\frac{E}{V} = \frac{N k_B T}{V}$$

$$\frac{\text{Joules}}{\text{m}^3} = \text{Pascals} = \text{Pressure}$$

⊛ Energy per volume behaves like pressure

$$P \sim \frac{N}{V} k_B T$$

$$P \sim \frac{n(N_A \cdot k_B) T}{V}$$

$$R = N_A k_B$$

$$P \sim \frac{nRT}{V}$$

$$\boxed{PV \sim nRT}$$

⊛ Recovered the ideal gas law from energy arguments. All based on kinetic theory and scaling not assumed from thermodynamics.

Laws of Thermodynamics

0th law $T_A = T_B$ and $T_B = T_C \Rightarrow T_A = T_C$

1st law $\Delta U = Q - W$
 $dU = \delta Q - \delta W$

2nd law $dS = \frac{\delta Q_{rev}}{T}$ $S = k_B \ln \Omega$ Entropy for equal-probability states (microcanonical)
 $\frac{dS}{dt} \gg 0$

3rd law $\lim_{T \rightarrow 0} S = 0$

Information (i) = $-\log p_i$

If $p_i p_j \Rightarrow -\log(p_i p_j) = -\log p_i - \log p_j$

Average Information = $\sum_i p_i \cdot (-\log p_i) \Rightarrow -\sum_i p_i \log p_i$

$S = -k_B \sum_i p_i \ln p_i$ Entropy is maximized when all microstates are equally likely: $p_i = \frac{1}{N}$ for all i

$S = -k_B \sum_{i=1}^N \frac{1}{N} \ln\left(\frac{1}{N}\right) = -k_B \cdot N \cdot \frac{1}{N} \ln\left(\frac{1}{N}\right) = k_B \ln N$

$\beta = \frac{1}{k_B T}$ $Z = \sum_j e^{-\beta E_j}$ Partition function $p_i = \frac{e^{-\beta E_i}}{Z}$ Boltzmann dist $\ln p_i = \ln\left(\frac{e^{-\beta E_i}}{Z}\right) = -\beta E_i - \ln Z$

$S = -k_B \sum_i p_i (-\beta E_i - \ln Z) = k_B \beta \sum_i p_i E_i + k_B \ln Z \sum_i p_i$

$\sum_i p_i = 1$

$\sum_i p_i E_i = \langle E \rangle$ (Average Energy)

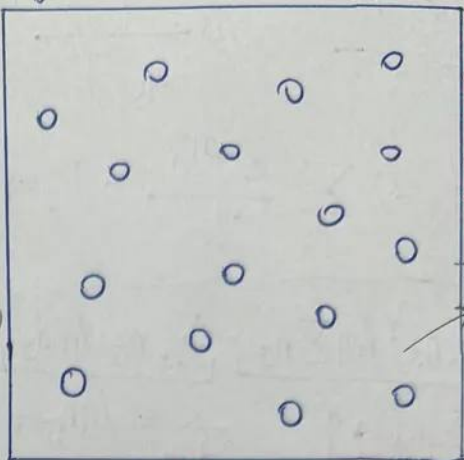
$S = k_B \beta \langle E \rangle + k_B \ln Z = k_B (\ln Z + \beta \langle E \rangle)$ Canonical Entropy Expression Derived from Gibbs Entropy

Energy Distribution in Molecular System

Microstate Count Formula $W = \frac{N!}{n_0! n_1! \dots n_M!}$

How many ways to permute N items grouped into $M+1$ categories.

$$\begin{matrix} \epsilon_0 & \epsilon_1 & \epsilon_2 & \dots & \epsilon_M \\ \uparrow & \uparrow & \uparrow & & \uparrow \\ n_0 & n_1 & n_2 & & n_M \end{matrix}$$



Degeneracy Function g

$$g_1 = \sum n_j - N = 0$$

$$g_2 = \sum n_j \epsilon_j - E = 0$$

constraint functions Use Lagrange multipliers to maximize $\ln W$ under these constraints to derive the Boltzmann distribution.

Full Derivation of Microstates Count W

$$W(\{n_j\}) = \frac{N!}{\prod_{j=0}^M n_j!}$$

Microcanonical Partition Function

$$\sum n_j = N$$

$$\Omega(E, N) = \sum_{\{n_j\}} \frac{N!}{\prod_{j=0}^M n_j!} \quad \text{Subject to } \sum n_j \epsilon_j = E$$

Constraints:

$$\sum_{j=0}^M n_j = N$$

how many particles are in energy level ϵ_j

$$\sum_{j=0}^M n_j \epsilon_j = E$$

Example: 4 particles across energy levels $\epsilon_0=0$, $\epsilon_1=1$, $\epsilon_2=2$ such that total energy is 4

	# particles at ϵ_0 n_0	# particles at ϵ_1 n_1	# particles at ϵ_2 n_2	Total Energy	$W = \frac{N!}{n_0! n_1! n_2!}$
0 th row	0	4	0	4	1
1 st row	1	2	1	4	12
2 nd row	2	0	2	4	6

Derivation of Boltzmann distribution via combinatorics and method of Lagrange multipliers, and using Stirling approximation $\ln n! \approx n \ln n - n$

$$\text{maximize } W(\{n_j\}) = \frac{(\sum n_j)!}{\prod n_j!} \equiv \text{maximize } \ln W = \ln \left(\frac{(\sum n_j)!}{\prod n_j!} \right) \approx \ln [(\sum n_j)!] - \sum_j \ln(n_j!)$$

$$\approx (\sum n_j) \ln(\sum n_j) - \sum n_j - \sum_j (n_j \ln n_j - n_j) \approx (\sum n_j) \ln(\sum n_j) - \sum_j n_j \ln n_j$$

Constraint 1: $g_1 = \sum n_j - N = 0$

Constraint 2: $g_2 = \sum n_j \epsilon_j - E = 0$

So we form Lagrangian

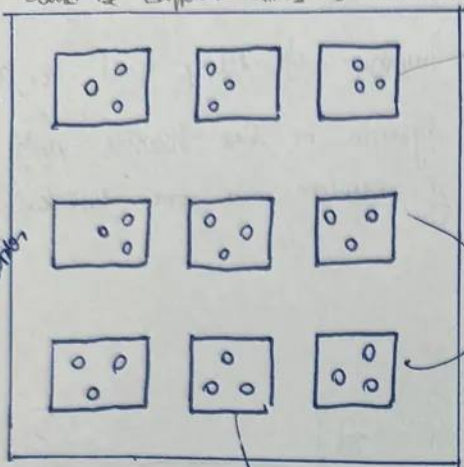
$$\mathcal{L} = \ln W - \lambda_1 g_1 - \lambda_2 g_2 \quad \text{then } \frac{\partial \mathcal{L}}{\partial n_j} = 0$$

Take derivative and solve for n_j :

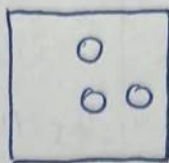
Microcanonical (NVE) Ensemble

Isolated. $N-V-E$ are fixed, all allowed microstates
Each system different microstate have equal probability
Could be different times or same time in different locations.

Example
Collection of states
@ diff time intervals
or
@ diff spatial locations



A states/systems



$S(N, V, E)$
↑
thermodynamic function
↓ ↓ ↓
control variables
 N, V, E

$$S(N, V, E) = k_B \ln(\Omega(N, V, E))$$

microstates
compatible with
 N, V, E

Assumption: All microstates are equally probable (since system is isolated)

First law of thermodynamics:

Change in internal energy = Heat added + Work done on system $dE = \delta Q + \delta W$

$$\delta Q = T dS \text{ (reversible)}$$

$$\delta W_{\text{mech}} = -P dV$$

$$dE = \delta Q + \delta W = T dS - P dV + \sum_I \mu_I dN_I$$

$$T = \left(\frac{\partial E}{\partial S} \right)_{V, \{N_I\}}$$

$$\delta W_{\text{chem}} = \sum_I \mu_I dN_I$$

$\frac{\partial E}{\partial S} = T$ Temperature is how much energy increases per unit entropy. (i.e., how much heat is needed to increase disorder)

$\frac{\partial E}{\partial V} = -P$ Pressure is the force pushing back when volume is compressed

$\frac{\partial E}{\partial N} = \mu$ Chemical potential is how energy changes when you add a particle

$$dE = T dS - P dV + \mu dN$$

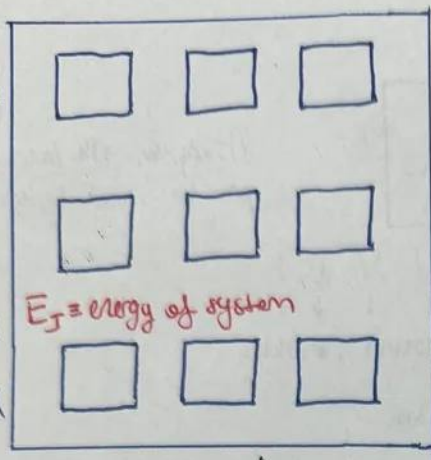
If I add heat, compress the system, or change particles - here's how energy responds.

Ensemble	You Fix	So You Want to Write -
NVE	S, V, N	$E(S, V, N) \Rightarrow dE = T dS - P dV + \mu dN$
NVT	T, V, N	We don't want dS , so use $F = E - TS \Rightarrow dF = -S dT - P dV + \mu dN$
NPT	T, P, N	Use $G = E - TS + PV \Rightarrow dG = -S dT + V dP + \mu dN$
μ VT	T, V, μ	Use $\Omega = E - TS - \mu N \Rightarrow d\Omega = -S dT - P dV - N d\mu$

Canonical Ensemble (NVT)

Fixed N, V, T
in contact with a
thermal reservoir
it can exchange
energy.

Energy fluctuates
between systems but
all share same N, V, T



Energy level of system

$E_1, E_2, \dots, E_\ell$

Occupation number of energy level $a_1, a_2, \dots, a_\ell$

a_J : number of systems in the ensemble with Energy E_J

Total number of systems in the ensemble is

$$\sum_{J=1}^{\ell} a_J = A$$

Microstate Counting for a Distribution:

$$W(\{a_J\}) = \frac{A!}{a_0! a_1! a_2! \dots a_\ell!} = \frac{(\sum a_J)!}{\prod a_J!}$$

A systems to energy levels $E_0, E_1, \dots, E_\ell$ such that a_J systems are in energy level E_J .

2) Probability of a Given Energy Level: $P_J = \frac{\bar{a}_J}{A}$ where $\bar{a}_J$: average # systems with energy E_J
 A : total # systems so P_J is the

probability that a random chosen system has energy E_J .

3) General property $\bar{M}$: Ensemble Average $\bar{M} = \sum_J M_J(E_J) \cdot P_J$

$M_J(E_J)$ value of the property if the system is in state E_J

P_J is the probability of being in that state.

4) How to compute $\bar{a}_J$ - Probabilistically

$$\bar{a}_J = \sum_{\{a_J\}} W(\{a_J\}) \cdot a_J(\{a_J\}) \cdot P(\{a_J\})$$

$W(\{a_J\})$: # ways to realize that occupation distribution

$a_J(\{a_J\})$: # ~~ways~~ systems in that configuration with energy E_J

$P(\{a_J\})$: probability of a configuration $\{a_J\}$ occurring

$$P_J = \frac{\bar{a}_J}{A} = \frac{1}{A} \sum_{\{a_J\}} W(\{a_J\}) \cdot a_J(\{a_J\}) \cdot P(\{a_J\})$$

Microscopic origin of Macroscopic probability of energy level E_J

$\{a_J\}$: one configuration of the ensemble

$W(\{a_J\})$: # ways to realize that configuration

$P(\{a_J\})$: probability of that configuration

$\bar{a}_J$: average # systems with Energy E_J

$P_J = \bar{a}_J / A$: probability that a randomly picked system is in energy E_J

$\bar{M} = \sum M_J P_J$: Ensemble average of any observable.

A : # systems in ensemble total

a_J : # systems in energy level E_J

$W(\{a_J\})$: # ways to arrange systems into this energy distribution.

P_J : Probability of finding a system with energy E_J

$\bar{M}$: Expected value of an observable property M .

Canonical Ensemble (NVT Ensemble)

$$p_i = \frac{e^{-\beta E_i}}{Z} \quad \beta = \frac{1}{k_B T} \quad Z = \sum_i e^{-\beta E_i}$$

$$\sum_i p_i = 1 \Rightarrow \sum_i \frac{e^{-\beta E_i}}{Z} = 1 \Rightarrow Z = \sum_i e^{-\beta E_i}$$

$$\ln W \approx \ln(\sum n_j) - \sum_j n_j \ln n_j \quad \frac{\partial \ln W}{\partial n_j} = \ln(\sum n_j) - \ln n_j$$

$$\frac{\partial}{\partial n_j} = \ln(\sum n_j) + 1 \quad \frac{\partial}{\partial n_j} = \ln n_j + 1$$

$$\text{now apply constraints: } \frac{\partial \ln W}{\partial n_j} - \lambda_1 \frac{\partial g_1}{\partial n_j} - \lambda_2 \frac{\partial g_2}{\partial n_j} = 0$$

$$\text{So final condition: } \ln(\sum n_j) - \ln n_j - \lambda_1 - \lambda_2 \epsilon_j = 0 \Rightarrow -\ln\left(\frac{n_j}{\sum n_j}\right) - \lambda_1 - \lambda_2 \epsilon_j = 0$$

$$\frac{n_j}{\sum n_j} = \exp(-\lambda_1 - \lambda_2 \epsilon_j) = \frac{\exp(-\lambda_2 \epsilon_j)}{\sum_j \exp(-\lambda_2 \epsilon_j)}$$

normalized probability, expressing the relative occupation of energy level ϵ_j .

Total particle constraint: $\sum_j n_j = N$

$$n_j = N \cdot \frac{\exp(-\lambda_2 \epsilon_j)}{\sum_j \exp(-\lambda_2 \epsilon_j)}$$

$$\text{Canonical partition function } Z = \sum_j \exp(-\lambda_2 \epsilon_j)$$

$$\text{then: } n_j = N \cdot \frac{\exp(-\lambda_2 \epsilon_j)}{Z}$$

$$\text{Link to physical temperature } \lambda_2 = \beta = \frac{1}{k_B T} \Rightarrow n_j \propto \exp\left(\frac{-\epsilon_j}{k_B T}\right)$$

$$n_j = N \cdot \frac{e^{-\epsilon_j/k_B T}}{Z} \quad \text{where } Z = \sum_j e^{-\epsilon_j/k_B T} \quad \text{Boltzmann Distribution}$$

$\frac{n_j}{\sum n_j}$: Probability of being in state j

$\exp(-\lambda_1 - \lambda_2 \epsilon_j)$: Unnormalized probability

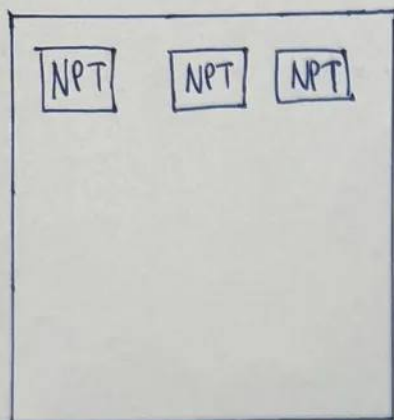
$\sum_j e^{-\lambda_2 \epsilon_j}$: Normalization constant

$e^{-\lambda_1}$: inverse of partition function $1/Z$

$\lambda_2 = \frac{1}{k_B T}$: Links stat mech to thermodynamics

$n_j \propto e^{-\epsilon_j/k_B T}$: Most probable distribution

Isothermal - Isobaric (NPT) Ensemble



A system

$a_{VJ} \equiv \#$ systems with volume V and energy E_{VJ}

$$W = \frac{A!}{\prod_V \prod_J a_{VJ}!} = \frac{(\sum_V \sum_J a_{VJ})!}{\prod_V \prod_J a_{VJ}!}$$

subject to

$$\begin{cases} \sum_V \sum_J a_{VJ} = A \\ g_1 \equiv \sum_V \sum_J a_{VJ} - A = 0 \\ \sum_V \sum_J a_{VJ} E_{VJ} = E \\ g_2 \equiv \sum_V \sum_J a_{VJ} E_{VJ} - E = 0 \\ \sum_V \sum_J a_{VJ} V = V \\ g_3 \equiv \sum_V \sum_J a_{VJ} V - V = 0 \end{cases}$$

~~$$\frac{\partial \ln W}{\partial a_{VJ}} - \alpha \frac{\partial g_1}{\partial a_{VJ}} - \beta \frac{\partial g_2}{\partial a_{VJ}} - \theta \frac{\partial g_3}{\partial a_{VJ}} = 0$$~~

$$\ln n! \approx n \ln n - n$$

$$\ln W = \ln \left(\frac{A!}{\prod_{V,J} a_{VJ}!} \right) = \ln A! - \ln (\prod_{V,J} a_{VJ}!) = A \ln A - A - (\sum_{V,J} a_{VJ} \ln a_{VJ} - \sum_{V,J} a_{VJ})$$

$$\ln W = (\sum a_{VJ}) \ln (\sum a_{VJ}) - \sum a_{VJ} \ln a_{VJ}$$

$$A = \sum_{V,J} a_{VJ} \quad \ln W = A \ln A - \sum_{V,J} a_{VJ} \ln a_{VJ} \quad A = \sum_{V,J} a_{VJ} \quad \frac{\partial A}{\partial a_{VJ}} = 1$$

$$\frac{\partial \ln W}{\partial a_{VJ}} = \frac{\partial}{\partial a_{VJ}} (A \ln A) - \frac{\partial}{\partial a_{VJ}} \left(\sum_{V,J} a_{VJ} \ln a_{VJ} \right) = \left(\frac{\partial}{\partial a_{VJ}} (A) \right) \ln A + A \left(\frac{\partial}{\partial a_{VJ}} (\ln A) \right) - (\ln a_{VJ} + 1)$$

$$\frac{\partial}{\partial a_{VJ}} \ln A = \frac{1}{A} \cdot \frac{\partial A}{\partial a_{VJ}} = \frac{1}{A}$$

$$\frac{\partial \ln W}{\partial a_{VJ}} = (\ln A + 1) - (\ln a_{VJ} + 1) = \ln A - \ln a_{VJ}$$

$$\frac{\partial \ln W}{\partial a_{VJ}} = \ln A - \ln a_{VJ}$$

$$\frac{\partial g_1}{\partial a_{VJ}} = 1$$

$$\frac{\partial g_2}{\partial a_{VJ}} = E_{VJ}$$

$$\frac{\partial g_3}{\partial a_{VJ}} = V$$

$$\frac{\partial \ln W}{\partial a_{VJ}} - \alpha \frac{\partial g_1}{\partial a_{VJ}} - \beta \frac{\partial g_2}{\partial a_{VJ}} - \theta \frac{\partial g_3}{\partial a_{VJ}} = 0$$

$$\ln A - \ln a_{VJ} - \alpha - \beta E_{VJ} - \theta V = 0$$

solve for a_{VJ} :

$$\ln a_{VJ} = \ln A - \alpha - \beta E_{VJ} - \theta V$$

$$a_{VJ} = A \cdot \exp(-\alpha - \beta E_{VJ} - \theta V) \quad a_{VJ} \propto \exp(-\beta E_{VJ} - \theta V)$$

$$P_{VJ} = \frac{a_{VJ}}{A} = \exp(-\alpha - \beta E_{VJ} - \theta V)$$

$$\exp(-\alpha) = \frac{1}{\sum_{V,J} \exp(-\beta E_{VJ} - \theta V)}$$

$$\sum_{V,J} P_{VJ} = 1 \quad \sum_{V,J} \exp(-\alpha - \beta E_{VJ} - \theta V) = 1 \quad \exp(-\alpha) \sum_{V,J} \exp(-\beta E_{VJ} - \theta V) = 1$$

$$P_{VJ} = \frac{\exp(-\beta E_{VJ} - \theta V)}{\sum_{V,J} \exp(-\beta E_{VJ} - \theta V)} \quad Z = \sum_{V,J} \exp(-\beta E_{VJ} - \theta V)$$

$$P_{VJ} = \frac{\exp(-\beta E_{VJ} - \theta V)}{Z} \quad \sum_{V,J} P_{VJ} = \frac{1}{Z} \sum_{V,J} \exp(-\beta E_{VJ} - \theta V) = 1$$